Control Systems (EEE/ECE/INSTR F242)

Tutorial 10 (24 March 2026)

1. For unity feedback closed loop system, the forward path transfer function G(s) is given as,

$$G(s) = K\frac{(s+1)(s+2)}{(s+0.1)(s-1)}$$

(A) Draw the root locus for this system.
(B) Find the gain 'K' for which the system will have repeated roots.
(C) Check if, $s_o = -1 + 0.678j$, lies on the root locus. If yes, find the gain which will cause this root.

2. Draw the root locus for a negative feedback system, with $G(s)H(s) = K/((s+1)^2(s+4))$. Find the value of K so that one of the closed loop poles is at -5.

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